CS410 Deep Learning · Coursework 2: Convolutional Networks

Deadline: Friday week 9, 16:00. Weight: 30% of the module.

Task
Implement and evaluate a small convolutional neural network for image classification.

Requirements
1. Write a MATLAB (or Python) script that builds a CNN with at least two convolutional layers.
2. Use same padding in every convolutional layer and justify the choice.
3. Add a pooling layer after each convolutional layer and state its effect on the feature map size.
4. Train the network and report training and validation accuracy per epoch.
5. Apply batch normalisation and compare results with and without it.
6. Write a 1500 word report containing: method, results with labelled plots, discussion, and conclusion.

Marking
Correct architecture and padding reasoning (30%). Experimental results (30%). Report quality (40%).

Submit the script and the report as a single zip on the module page.